Monitoring and alerting

AWS Academy is the main lesson. This short deck connects it to our class work.

PRIMARY

AWS Academy

Cloud Security Foundations · Module 6

IN CLASS

Practice

Lab 6.1 · CloudTrail / CloudWatch (75 min)

COURSE WORK

Local task

Local 09 · C5R1 Part 2 / C5R2 Part 1

Local 09 due 1 Mar · Use only your assigned Part

One path through the topic

STEP 1

Record

CloudTrail records the relevant API change.

STEP 2

Detect

EventBridge or a CloudWatch metric filter matches the event.

STEP 3

Notify

SNS delivery needs a confirmed subscription; inspect alarm and log evidence.

Does a CloudTrail trail by itself send an email alert?

What to do this week

BEFORE CLASS

Prepare

Open Module 6 and confirm access to an email inbox for the SNS step.

IN CLASS

Work

Complete Lab 6.1; capture the event, rule and alert evidence.

AFTER CLASS

Record

Use the Local 09 slides and lab outside class; submit by 1 Mar.

AWS work stays in AWS Academy · Local work stays in Classroom

Outside the Week 17 class

LOCAL 09

Apply the detector model

Study slides 5–7, then complete only your assigned Part outside class.

MODULE 8

Bridge work

Use the assigned LMS material to finish the security course sequence.

AWS ACADEMY

Course Assessment

Take it under LMS instructions; check the deadline in Classroom.

Local 09 due 1 Mar; late submissions close 8 Mar at 23:59.

Choose what the detector counts

PAIR

IP + username

A threshold on each pair can miss a spray that tries many usernames once from one IP.

SPRAY

Group by source

The local detector groups by source IP to see many usernames from the same origin.

BRUTE FORCE

Group by username

Counting attempts for one username detects a different pattern. Test both patterns against the rule.

Local 09 simulates grouping; avoid high-cardinality CloudWatch metric dimensions in production.

Two ways to turn an event into an alert

EVENT ROUTE

CloudTrail → EventBridge

A rule matches a relevant event and sends it to a target such as SNS.

METRIC ROUTE

Logs → metric → alarm

A CloudWatch Logs filter counts matching events; an alarm evaluates the metric and can notify.

COMPARE

Separate paths

CloudTrail records the action. EventBridge and a metric alarm need not be one serial pipeline.

Use the AWS lab for the real service setup; Local 09 studies the counting choice.

The same events, different counts

SPRAY

20 names · one IP

One failure per name: pair and username counters stay at 1; a source-IP counter reaches 20.

DISTRIBUTED GUESSES

One name · 20 IPs

One failure per IP: pair and source-IP counters stay at 1; a username counter reaches 20.

SCOPE

A simulation

The local app resets a streak on success. A real metric filter counts matching events in an alarm window.

A running detector can miss an attack when its grouping key does not match the threat.